

A LINEAR ALGORITHM FOR HUANG’S QUADRATIC FORM ON NUMERICAL SEMIGROUPS AND DENSITY ASYMPTOTICS

ARTUR FLAMANDZKI

ABSTRACT. Let $G = \mathbb{N} \setminus \langle a, b \rangle$ be the gap set of the numerical semigroup generated by coprime $a < b$, and $N = |G|$. Huang [1] defined the quadratic form $Q(\mathbf{n}) = \sum_{i,j} K(j-i) n_i n_j$ on $\mathbb{R}^G$, where $K(d) = \mathbf{1}_{d \geq 0} - \mathbf{1}_{d \geq a} - \mathbf{1}_{d \geq b} + \mathbf{1}_{d \geq a+b}$, and showed that it recovers the dinv statistic on rational Dyck paths. The naive evaluation of Q requires $O(N^2)$ operations.

We prove that the interval structure of K allows one to reduce Q to a linear combination of sliding-window sums, yielding an $O(N)$ algorithm (Theorem 1.1). We generalise to semigroups $\langle p_1, \dots, p_n \rangle$ with n generators, where the generalised kernel $K^{(n)}$ defined by inclusion–exclusion — with 2^n terms — has at most $w(n) \leq \sigma_n$ active windows due to parity cancellation; these are computable incrementally in time $O(n \cdot \sigma_n)$ (Theorem 1.2). Empirically $w(n) \sim \sigma_n$ while $w(n) \ll 2^n$: exponential combinatorics reduces to a sparse interval structure. For a sequence of pairwise distinct integers $p_i \geq 2$ satisfying $\log p_n = o(n)$ we prove

$$\frac{w(n)}{2^n - 1} = o\left(\prod_{i=1}^n \left(1 - \frac{1}{p_i}\right)\right), \quad n \rightarrow \infty$$

(Theorem 1.5).

1. INTRODUCTION

1.1. Background and motivation. Let $a, b \in \mathbb{N}$, $\gcd(a, b) = 1$, $a < b$. The numerical semigroup generated by a and b is $\langle a, b \rangle = \{xa + yb : x, y \in \mathbb{Z}_{\geq 0}\}$. By the Sylvester–Frobenius theorem the gap set $G = \mathbb{N} \setminus \langle a, b \rangle$ is finite, $|G| = N = \frac{(a-1)(b-1)}{2}$. We write $G = \{g_0 < g_1 < \dots < g_{N-1}\}$.

Huang [1] defined on $\mathbb{R}^G$ the quadratic form

$$Q(\mathbf{n}) = \sum_{i,j \in G} K(j-i) n_i n_j, \quad K(d) = \mathbf{1}_{d \geq 0} - \mathbf{1}_{d \geq a} - \mathbf{1}_{d \geq b} + \mathbf{1}_{d \geq a+b},$$

and showed that when $\mathbf{n} = \mathbf{1}_D$ is the indicator vector of a Young subdiagram $D \subseteq G$, then $Q(\mathbf{1}_D) = \text{dinv}(D)$, where dinv is the Gorsky–Mazin statistic on rational Dyck paths. An independent formal verification of this theorem in Lean 4/Mathlib is given in [2].

The paper [1] and its formalisation [2] concern the identity $Q(\mathbf{1}_D) = \text{dinv}(D)$. The present paper concerns the computational efficiency of Q and its generalisation to n generators.

Direct evaluation of Q from its definition requires N^2 operations. The key observation is that K takes at most three nonzero values on five disjoint intervals (Lemma 2.1), which reduces the computation of Q to two sliding-window sums, giving an $O(N)$ algorithm.

The generalisation to n generators defines the kernel $K^{(n)}$ by inclusion–exclusion over all 2^n subsets of $\{p_1, \dots, p_n\}$. The interval structure of $K^{(n)}$ is preserved, but the number of active windows $w(n)$ — intervals on which $K^{(n)} \neq 0$ — is far smaller than 2^n due to parity cancellation. We prove $w(n) \leq \sigma_n$ and give an incremental algorithm computing $w(n)$ and the active windows themselves.

The ratio $w(n)/(2^n - 1)$ is naturally linked to the product $P(n) = \prod_{i=1}^n (1 - 1/p_i)$: under the condition $\log p_n = o(n)$ we show $w(n)/(2^n - 1) = o(P(n))$, i.e., the density of active windows is asymptotically negligible relative to $P(n)$.

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E-mail: flamandzki.artur@gmail.com.

The generalisation to n generators does not address the Frobenius problem, which asks for an explicit formula for the largest element of $\mathbb{Z}_{\geq 0} \setminus \langle p_1, \dots, p_n \rangle$ and remains open for $n \geq 3$ [3]. The kernel $K^{(n)}$ is defined by inclusion–exclusion over the generator set; the algorithm takes G as input.

1.2. Definitions and main results. Two generators. For fixed k and $\mathbf{n} = (n_j)_{j \in G} \in \mathbb{R}^G$ define the window sums

$$W_k^+ := \sum_{\substack{j \in G \\ g_k \leq g_j < g_k + a}} n_j, \quad W_k^- := \sum_{\substack{j \in G \\ g_k + b \leq g_j < g_k + a + b}} n_j.$$

Theorem 1.1 (Linear algorithm, two generators). *For every $\mathbf{n} \in \mathbb{R}^G$,*

$$Q(\mathbf{n}) = \sum_{k=0}^{N-1} n_k (W_k^+ - W_k^-).$$

Both families (W_k^+) and (W_k^-) are jointly computable in time $O(N)$ and space $O(N)$ using four sliding-window pointers on G .

Theorem 1.1 reduces the cost from $O(N^2)$ to $O(N)$. Since $N = \frac{(a-1)(b-1)}{2}$, the algorithm runs in time $O(ab)$ versus $O(a^2b^2)$ for the naive approach.

Multiple generators. Let $\mathbf{p} = (p_1, \dots, p_n)$, $p_i \geq 2$ pairwise distinct, $\gcd(p_1, \dots, p_n) = 1$, $\sigma_n = \sum_{i=1}^n p_i$. The generalised kernel

$$K^{(n)}(d) := \sum_{S \subseteq \{p_1, \dots, p_n\}} (-1)^{|S|} \mathbf{1}_{d \geq \sigma(S)}, \quad \sigma(S) = \sum_{p \in S} p, \quad \sigma(\emptyset) = 0,$$

is nonzero on at most $w(n)$ intervals between consecutive breakpoints; we write $w(n)$ for the number of active windows.

Theorem 1.2 (Sliding-window algorithm, n generators). *Let $G = \mathbb{N} \setminus \langle p_1, \dots, p_n \rangle$. The form $Q(\mathbf{n}) = \sum_{i,j \in G} K^{(n)}(j-i) n_i n_j$ is computable in time $O(T_{\text{prep}} + w(n) \cdot N)$, where $T_{\text{prep}} = O(n \cdot \sigma_n)$. One has $w(n) \leq \sigma_n$. The set of active windows and the values of $K^{(n)}$ on them are computable incrementally: the transition from $n-1$ to n generators costs $O(\sigma_{n-1})$.*

Density asymptotics. For a sequence of pairwise distinct integers $p_i \geq 2$ define the density product and density defect:

$$P(n) := \prod_{i=1}^n \left(1 - \frac{1}{p_i}\right), \quad \delta(n) := P(n) - \frac{w(n)}{2^n - 1}.$$

Remark 1.3. The term *gap* refers exclusively to an element of $G = \mathbb{N} \setminus \langle p_1, \dots, p_n \rangle$. The symbol $\delta(n)$ refers exclusively to the quantity $P(n) - w(n)/(2^n - 1)$. The two notions are independent.

Remark 1.4. The product $P(n) = \prod_{i=1}^n (1 - 1/p_i)$ is defined for any sequence of pairwise distinct integers $p_i \geq 2$ and does not require the p_i to be prime. When the p_i are consecutive primes, $P(n)$ coincides with the classical Mertens product; in that case $P(n) \rightarrow 0$.

Theorem 1.5 (Density asymptotics). *Let $(p_i)_{i \geq 1}$ be a sequence of pairwise distinct integers ≥ 2 satisfying $\log p_n = o(n)$. Then*

$$\frac{w(n)}{2^n - 1} = o(P(n)), \quad n \rightarrow \infty,$$

equivalently, $\delta(n)/P(n) \rightarrow 1$.

The condition $\log p_n = o(n)$ is satisfied by any family of prime generators ($p_n \sim n \log n$), twin primes, arithmetic progressions of primes, and sequences of random odd numbers of prime density.

2. PRELIMINARIES

Two-generator kernel.

Lemma 2.1 (Interval structure of K). *For all $d \in \mathbb{Z}$:*

$$K(d) = \begin{cases} 0 & d < 0, \\ +1 & 0 \leq d < a, \\ 0 & a \leq d < b, \\ -1 & b \leq d < a+b, \\ 0 & d \geq a+b. \end{cases}$$

Proof. Direct inspection of the four indicators $\mathbf{1}_{d \geq 0}$, $-\mathbf{1}_{d \geq a}$, $-\mathbf{1}_{d \geq b}$, $+\mathbf{1}_{d \geq a+b}$ for $a < b$. The five cases are mutually exclusive and exhaustive. $\square$

Corollary 2.2. $K(d) \neq 0$ implies $0 \leq d < a+b$. The sum $\sum_{d \in \mathbb{Z}} K(d) = a - (b - a) = 2a - b$.

Multi-generator kernel.

Definition 2.3 (Parity function). For $\mathbf{p} = (p_1, \dots, p_n)$ and $s \in \mathbb{Z}_{\geq 0}$:

$$\pi_n(s) := \sum_{\substack{S \subseteq \{p_1, \dots, p_n\} \\ \sigma(S) = s}} (-1)^{|S|}.$$

One has $K^{(n)}(d) = \sum_{s \leq d} \pi_n(s)$, so $K^{(n)}$ is constant on intervals between consecutive s with $\pi_n(s) \neq 0$.

Lemma 2.4 (Vanishing outside $[0, \sigma_n]$). $K^{(n)}(d) = 0$ for $d < 0$ and for $d \geq \sigma_n$.

Proof. For $d < 0$: every indicator $\mathbf{1}_{d \geq \sigma(S)} = 0$. For $d \geq \sigma_n$: all indicators equal 1, and $\sum_{S \subseteq \{p_1, \dots, p_n\}} (-1)^{|S|} = (1-1)^n = 0$ by the binomial theorem. $\square$

Lemma 2.5 (Parity recurrence).

$$\pi_{n+1}(s) = \pi_n(s) - \pi_n(s - p_{n+1}).$$

Proof. Subsets of $\{p_1, \dots, p_{n+1}\}$ with sum s split into: subsets $S \subseteq \{p_1, \dots, p_n\}$ with $\sigma(S) = s$, contributing $\pi_n(s)$, and subsets $S \cup \{p_{n+1}\}$ with $\sigma(S) = s - p_{n+1}$, contributing $-\pi_n(s - p_{n+1})$. $\square$

3. PROOF OF THEOREM 1.1

3.1. Reduction of the quadratic form.

Lemma 3.1. *For fixed $i \in G$:*

$$\sum_{j \in G} K(j - i) n_j = W_i^+ - W_i^-.$$

Proof. By Lemma 2.1: $K(j - i) = +1$ when $g_i \leq g_j < g_i + a$, $K(j - i) = -1$ when $g_i + b \leq g_j < g_i + a + b$, and $K(j - i) = 0$ otherwise. The definitions of W_i^+ and W_i^- correspond precisely to these two intervals. $\square$

Proof of Theorem 1.1. Rearranging the sum:

$$Q(\mathbf{n}) = \sum_{i \in G} n_i \sum_{j \in G} K(j - i) n_j = \sum_{i \in G} n_i (W_i^+ - W_i^-),$$

where the first equality follows by symmetry (index swap) and the second from Lemma 3.1.

Linearity: maintain four pointers $\text{lo}^+, \text{hi}^+, \text{lo}^-, \text{hi}^-$ into sorted G , initialised to 0. For $k = 0, 1, \dots, N-1$: advance hi^+ while $g_{\text{hi}^+} < g_k + a$; advance lo^+ while $g_{\text{lo}^+} < g_k$; advance hi^- while $g_{\text{hi}^-} < g_k + a + b$; advance lo^- while $g_{\text{lo}^-} < g_k + b$. Each pointer is non-decreasing and traverses G at most once: at most $4N$ updates in total. $\square$

Corollary 3.2. *For $(a, b) = (97, 101)$, $N = 4801$: the $O(N)$ algorithm performs 19 204 operations versus 23 049 601 for $O(N^2)$; ratio ≈ 1200 . In general $N^2/(4N) = N/4 = (a - 1)(b - 1)/8$.*

4. PROOF OF THEOREM 1.2

4.1. Active windows and the bound $w(n) \leq \sigma_n$.

Definition 4.1 (Active window). An interval $[s, s']$ between consecutive breakpoints of $K^{(n)}$ is *active* if $K^{(n)} \neq 0$ on (s, s') . $w(n)$ denotes the number of active windows.

Lemma 4.2. $w(n) \leq \sigma_n$.

Proof. By Lemma 2.4 the support of $K^{(n)}$ is contained in $[0, \sigma_n) \cap \mathbb{Z}$, which contains σ_n integers. The number of active windows does not exceed the number of breakpoints, which is at most σ_n . $\square$

Lemma 4.3 (Parity cancellation). *If $\pi_n(s) = 0$, then s is not a breakpoint of $K^{(n)}$, reducing $w(n)$ below σ_n . This occurs when there exist $S_1 \neq S_2$ with $\sigma(S_1) = \sigma(S_2) = s$ and $|S_1| \not\equiv |S_2| \pmod{2}$.*

Proof. By the definition of π_n : contributions of opposite parity cancel. $\square$

Remark 4.4. The sparsity ratio $1 - w(n)/\sigma_n$ grows from 0% at $n = 2$ to 100.0% at $n \geq 21$ for the sequence of the first primes (Table 3; values 100.0% result from rounding to one decimal place). A combinatorial proof of the strict inequality $w(n) < 2^n - 1$ for every $n \geq 3$ remains open (Section 6, item (iii)).

4.2. Incremental algorithm.

Lemma 4.5 (Window update). *The active windows for $n + 1$ generators are determined by the new breakpoints arising from $\pi_{n+1}(s) = \pi_n(s) - \pi_n(s - p_{n+1})$ (Lemma 2.5). The transition from n to $n + 1$ generators costs $O(\sigma_n)$.*

Proof. The support of π_n is contained in $[0, \sigma_n]$, so the update $\pi_{n+1}(s) = \pi_n(s) - \pi_n(s - p_{n+1})$ requires at most $\sigma_n + 1$ operations. $\square$

Proof of Theorem 1.2. We compute $\mathcal{W}$ — the set of triples (lo, hi, c) representing active windows with values of $K^{(n)}$ — by the incremental algorithm: start from $n = 1$ (one window $[0, p_1)$ with value +1) and perform $n - 1$ steps of Lemma 4.5. Total cost: $\sum_{k=1}^{n-1} O(\sigma_k) = O(n \cdot \sigma_n)$.

Then for each window $(lo, hi, c) \in \mathcal{W}$ perform one sliding-window pass over G in time $O(N)$: accumulate $c \cdot \sum_{j \in G \cap [lo, hi)} n_j$. Total cost: $O(T_{\text{prep}} + w(n) \cdot N)$. $\square$

Remark 4.6. The change in $K^{(n)}$ upon adding generator p_{n+1} is

$$K^{(n+1)}(d) - K^{(n)}(d) = \sum_{\substack{S \subseteq \{p_1, \dots, p_n\} \\ \sigma(S) + p_{n+1} \leq d}} (-1)^{|S|+1}.$$

A position inactive under $K^{(n)}$ becomes active under $K^{(n+1)}$ if and only if this sum is nonzero. Empirically $K^{(n)}(d) \notin \{-1, 0, 1\}$ is possible; values ± 3 are observed at $n = 12$ (Table 4).

5. PROOF OF THEOREM 1.5

Proof of Theorem 1.5. We show that $w(n)/((2^n - 1)P(n)) \rightarrow 0$.

Step 1. By Lemma 4.2: $w(n) \leq \sigma_n \leq np_n$.

Step 2. Since $p_1 < p_2 < \dots < p_n$ are pairwise distinct and ≥ 2 , we have $p_k \geq k + 1$ for each k , so

$$H_n := \sum_{k=1}^n \frac{1}{p_k} \leq \sum_{j=2}^{n+1} \frac{1}{j} \leq \log(n + 1).$$

From the inequality $\log(1 - x) \geq -2x$ for $x \in [0, 1/2]$:

$$\log P(n) = \sum_{k=1}^n \log\left(1 - \frac{1}{p_k}\right) \geq -2H_n \geq -2\log(n+1),$$

whence $P(n) \geq (n+1)^{-2}$.

Step 3. From Steps 1–2 and $2^n - 1 \geq 2^{n-1}$:

$$\frac{w(n)}{(2^n - 1)P(n)} \leq \frac{2np_n(n+1)^2}{2^n}.$$

Step 4. By the hypothesis $\log p_n = o(n)$:

$$\log \frac{p_n}{2^n} = \log p_n - n \log 2 = o(n) - n \log 2 \rightarrow -\infty,$$

so $p_n/2^n \rightarrow 0$, while the factor $2n(n+1)^2$ grows polynomially. Hence the expression in Step 3 tends to 0, which is equivalent to $\delta(n)/P(n) \rightarrow 1$. $\square$

Remark 5.1. For $n \geq 25$ in Table 3: $w(n)/(2^n - 1) < 10^{-4}$, i.e., $w(n)/(2^n - 1)$ is indistinguishable from zero at four decimal places of $P(n)$ across all eight tested generator families.

6. OPEN PROBLEMS

- (i) **Boundedness of $|K^{(n)}(d)|$.** Does $\sup_{n,d} |K^{(n)}(d)| < \infty$? Empirically $|K^{(n)}(d)| \leq 3$ for $n \leq 12$ (Table 4).
- (ii) **Stability of values.** For each fixed d , does there exist $n_0 = n_0(d)$ such that $K^{(n)}(d) = K^{(n_0)}(d)$ for all $n \geq n_0$?
- (iii) **Sharp inequality $w(n) < 2^n - 1$.** Theorem 1.5 proves $w(n)/(2^n - 1) \rightarrow 0$, but does not give a sharp bound. Does $w(n) < 2^n - 1$ hold for every $n \geq 3$ and every sequence of coprime generators?
- (iv) **Characterisation of n^* .** Let $n^* = \min\{n : \delta(n) > 0\}$. Theorem 1.5 proves $\delta(n)/P(n) \rightarrow 1$, but does not characterise n^* . Conjecture 6.1 suggests that n^* depends only on the harmonic sum H_n .
- (v) **Monotone behaviour of $\delta(n)$.** Empirically $\delta(n)$ attains its maximum at $n = 16$ and then decreases (Table 3). Does $\delta(n)$ attain a proper maximum at finite n ?
- (vi) **Structure of $\Delta w(n)$.** From $n = 19$ onwards: the increments $\Delta w(n) = w(n) - w(n-1)$ form two increasing subsequences according to the parity of n . What is the arithmetic mechanism behind this split?
- (vii) **Asymptotics of $\Delta w(n)$.** Does $\Delta w(n)/p_n \rightarrow 1$ for sequences with $p_n \rightarrow \infty$? Numerical data: $p_{300} = 1987$, $\Delta w(300) = 1986$ (baseline sequence); $p_{300} = 2153$, $\Delta w(300) = 2153$ (sequence starting from 101). The convergence $\Delta w(n) \sim p_n$ would imply $w(n) \sim \sigma_n$.

Conjecture 6.1. *Let $n^* = \min\{n : \delta(n) > 0\}$. The value n^* depends only on the harmonic sum $H_n = \sum_{k=1}^n 1/p_k$: sequences with the same H_n as the first n primes share the same n^* , regardless of the arithmetic nature of the generators.*

The conjecture is confirmed experimentally (Table 5). A formal proof requires the characterisation of n^* described in problem (iv).

Related literature. Parity cancellation in inclusion–exclusion appears in unrelated contexts: in the theory of chromatic polynomials (Whitney [4]; Chen and Qian [5]) and in analytic sieve theory (Tao [6]). The interval sparsity of $K^{(n)}$ and parity cancellation for numerical semigroup kernels have not, to our knowledge, been described previously.

7. CONCLUSIONS

The starting point is Huang’s observation [1]: the quadratic form Q on the gap set of the numerical semigroup $\langle a, b \rangle$ recovers the combinatorial statistic dinv . The present paper addresses the computational efficiency of Q and reveals a structure that has not been described before.

Complexity reduction. The interval structure of K (Lemma 2.1) allows one to replace the double sum $O(N^2)$ by two sliding-window passes in time $O(N)$ (Theorem 1.1). With n generators the kernel $K^{(n)}$ is defined by 2^n inclusion–exclusion terms, but due to parity cancellation it has at most $w(n) \leq \sigma_n$ active windows (Theorem 1.2). Empirically $w(n) \sim \sigma_n$ while $w(n) \ll 2^n$: exponential combinatorics reduces to a sparse interval structure, and the incremental algorithm determines this structure in time $O(n \cdot \sigma_n)$.

Asymptotics. Theorem 1.5 quantifies this phenomenon: under the subexponential condition $\log p_n = o(n)$,

$$\frac{w(n)}{2^n - 1} = o\left(\prod_{i=1}^n \left(1 - \frac{1}{p_i}\right)\right), \quad n \rightarrow \infty.$$

The density of active windows is asymptotically negligible relative to the density product $P(n)$; equivalently $\delta(n)/P(n) \rightarrow 1$. Numerical data (Table 3) confirm the convergence for $n \leq 30$ and indicate a maximum of $\delta(n)$ at $n = 16$, whose combinatorial characterisation remains open.

Outlook. Parity cancellation in $K^{(n)}$ admits an algebraic formulation via the difference operator $\Delta_p f(d) = f(d) - f(d - p)$ acting on the indicator function $\mathbf{1}_{[0, \infty)}$:

$$K^{(n)} = \Delta_{p_1} \circ \cdots \circ \Delta_{p_n} \mathbf{1}_{[0, \infty)}.$$

In this formulation $\pi_n(s)$ is the coefficient of the operator mask, and the sparsity of $K^{(n)}$ corresponds to zero coefficients of that mask. The connection with generating functions, operator algebra, and harmonic analysis on $\mathbb{Z}$ points to directions for further research.

APPENDIX A. NUMERICAL VERIFICATION

A.1. EXPERIMENTAL CONDITIONS

Experiments were conducted in Python 3.14 (SymPy 1.14, NumPy) on a workstation AMD Ryzen 9 5900X (64 GB, Windows 11 Pro) and a laptop Intel Core i7-13620H (16 GB, Windows 11 Pro). Arithmetic: IEEE 754 double precision ($\varepsilon \approx 2.2 \times 10^{-16}$).

Three implementations — naive $O(N^2)$, prefix sums $O(N \log N)$, and sliding window $O(N)$ — form a verification chain. Operation counts are deterministic functions of (a, b) . Cross-checking by direct enumeration $O(2^n)$ at $n = 27, 28$ was performed multithreaded; all remaining integer values ($w(n)$, operation counts, active positions) were bitwise identical on both platforms. All verification scripts are available at https://github.com/Flaman55/RelMathApps/tree/main/number_theory/numerical_semigroups/linear_Q: `verify_linear_Q.py` (Theorems 1.1–1.2, Tables 1–2), `appendix_verify.py` (Tables 3–5), `incremental_windows.py` (incremental algorithm), and `section_C_parallel.py` (parallel enumeration, $n = 19, \dots, 30$).

A.2. TWO GENERATORS

a	b	N	ops. $O(N^2)$	ops. $O(N)$	max. error
3	5	4	16	16	5.3×10^{-15}
5	7	12	144	48	1.8×10^{-14}
7	11	31	961	124	1.1×10^{-13}
11	13	60	3600	240	2.8×10^{-13}
23	29	314	98596	1256	1.2×10^{-11}
37	41	721	519841	2884	4.3×10^{-11}
97	101	4801	23049601	19204	1.3×10^{-9}

All 450 pairs (a, b) with $b \leq 40$: **PASS**. Tolerance 10^{-9} (10^{-8} for $N > 500$).

TABLE 1. Verification of Theorem 1.1: two generators. Operation ratio $N^2/(4N) = N/4$.

A.3. MULTIPLE GENERATORS

Generators	N	$w(n)$	max. error	Status
$\{2, 3\}$	1	2	0	OK
$\{3, 5\}$	4	2	8.9×10^{-16}	OK
$\{2, 3, 5\}$	1	4	0	OK
$\{3, 5, 7\}$	3	5	1.8×10^{-15}	OK
$\{2, 3, 5, 7\}$	1	8	0	OK

TABLE 2. Verification of Theorem 1.2: multiple generators against naive $O(N^2)$.

A.4. DENSITY DATA (THEOREM 1.5)

n	$2^n - 1$	$w(n)$	$P(n)$	Sparsity%	$\delta(n)$
2	3	2	0.3333	0.0%	-0.3333
3	7	4	0.2667	14.3%	-0.3048
4	15	8	0.2286	26.7%	-0.3048
5	31	14	0.2078	29.0%	-0.2438
6	63	20	0.1918	44.4%	-0.1257
7	127	30	0.1805	59.1%	-0.0557
8	255	38	0.1710	72.2%	+0.0220
9	511	48	0.1636	81.6%	+0.0697
10	1023	60	0.1579	88.0%	+0.0993
11	2047	84	0.1529	92.5%	+0.1118
12	4095	116	0.1487	95.3%	+0.1204
13	8191	146	0.1451	97.2%	+0.1273
14	16383	198	0.1417	98.3%	+0.1296
15	32767	236	0.1387	99.0%	+0.1315
16	65535	288	0.1361	99.4%	+0.1317
17	131071	352	0.1338	99.7%	+0.1311
18	262143	412	0.1316	99.8%	+0.1300
19	524287	492	0.1296	99.9%	+0.1287
20	1048575	564	0.1278	99.9%	+0.1273
21	2097151	646	0.1260	100.0%	+0.1257
22	4194303	720	0.1245	100.0%	+0.1243
23	8388607	808	0.1230	100.0%	+0.1229
24	16777215	892	0.1216	100.0%	+0.1215
25	33554431	996	0.1203	100.0%	+0.1203
26	67108863	1096	0.1191	100.0%	+0.1191
27	134217727	1200	0.1180	100.0%	+0.1180
28	268435455	1306	0.1169	100.0%	+0.1169
29	536870911	1416	0.1158	100.0%	+0.1158
30	1073741823	1528	0.1148	100.0%	+0.1148

TABLE 3. Number of active windows $w(n)$, density product $P(n)$, sparsity $1 - w(n)/\sigma_n$, and density defect $\delta(n)$ (Theorem 1.5) for $p_i = i$ -th prime. Milestones: $n = 8$ — first $\delta(n) > 0$; $n = 21$ — sparsity reaches 100% (after rounding); $n = 25$ — $w(n)/(2^n - 1) < 10^{-4}$. Maximum of $\delta(n)$ at $n = 16$ (bold). Values “100.0%” result from rounding to one decimal place. Data $n \leq 18$: direct enumeration; $n = 19, \dots, 30$: incremental algorithm, independently confirmed for $n = 27, 28$ by $O(2^n)$ enumeration.

A.5. VALUES OF $K^{(n)}(d)$ — ACTIVATION AND DEACTIVATION

d	$n=2$	$n=3$	$n=4$	$n=5$	$n=6$	$n=7$	$n=8$	$n=9$	$n=10$	$n=11$	$n=12$
0	1	1	1	1	1	1	1	1	1	1	1
3	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1
5	.	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1
7	.	.	-1	-1	-1	-1	-1	-1	-1	-1	-1
8	.	1	.	.	.	.	.	.	.	.	.
9	.	.	1	1	1	1	1	1	1	1	1
10	.	.	1	1	1	1	1	1	1	1	1
16	.	.	.	.	1	1	1	1	1	1	1
17	.	.	.	1	2	1	1	1	1	1	1
25	.	.	.	.	.	.	1	1	1	1	1
47	.	.	.	.	.	1	1	.	-1	-2	-3

TABLE 4. Selected values of $K^{(n)}(d)$ for $n = 2, \dots, 12$ ($p_i = i$ -th prime). Dot ($\cdot$): $K^{(n)}(d) = 0$. Position $d = 8$: active at $n = 3$, deactivated at $n = 4$. Position $d = 17$: $K^{(6)}(17) = 2$. Position $d = 47$: $K^{(12)}(47) = -3$. The table provides empirical examples of values $|K^{(n)}(d)| = 2, 3$, motivating open problem (i). The phenomenon of position activation and deactivation, empirically arising from parity cancellation among overlapping subset-sum representations, is the subject of open problem (ii).

A.6. VERIFICATION OF CONJECTURE 6.1

Generator sequence	First gen.	n^*	$\delta/P, n = 25$	$\delta/P, n = 300$
<i>Prime sequences</i>				
First n primes	2	8	0.9998	1.0000
Primes from 101	101	2	0.9999	1.0000
Primes from 1009	1009	2	0.9997	1.0000
Every other prime ($p_2, p_4, \dots$)	3	8	0.9998	1.0000
Every third prime ($p_3, p_6, \dots$)	5	2	0.9998	1.0000
Smaller of twin prime pairs	3	9	0.9996	1.0000
Primes $\equiv 3 \pmod{4}$	3	8	0.9998	1.0000
Primes $\equiv 1 \pmod{4}$	5	2	0.9998	1.0000
<i>Random sequences (random odd integers)</i>				
Same H_n as baseline (seed 42)	—	8	0.9999	—
Same H_n as baseline (seed 137)	—	8	0.9999	—
Different H_n (seed 42)	—	2	0.9999	—

TABLE 5. Data consistent with Theorem 1.5: $\delta(n)/P(n) \rightarrow 1$ for all tested sequences. At $n = 300$: $\delta(n)/P(n) = 1.0000$ (six decimal places) for all prime sequences. Random sequences with the same H_n as the baseline yield identical n^* , consistent with Conjecture 6.1.

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